GATE DA 2027

Linear Algebra

Topic-wise Question Bank

Weighted by what GATE DA 2024–2026 actually asked

202 questions • **10** topics

MCQ • MSQ • NAT — the three GATE question types

Answer key at the end — answers only, no worked solutions

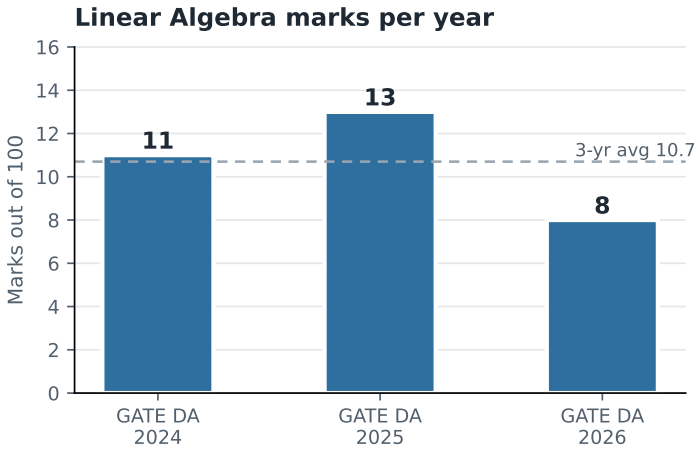
Every numerical answer in this book was recomputed independently in Python before the key was fixed — all 105 machine checks agreed. Options are shuffled, so the key is not all "A".

Prepared for Ashutosh • August 2026

How this book is built

Linear Algebra is a steady 8 to 13 marks. Reading the official GATE DA question papers directly, Linear Algebra carried about **11 marks in 2024** (7 questions) and **13 marks in 2025** (8 questions); published analyses put **2026 at about 8 marks**. The average is close to **11**, and the sensible planning range for 2027 is **8 to 15 marks**.

These counts are ±2. Questions that sit on a boundary — a PCA question, a Gram-matrix question, a covariance-matrix question — get assigned to Machine Learning by some analysts and to Linear Algebra by others. The 2024 and 2025 figures here come from reading the official papers; the 2026 figure comes from a published analysis.



What GATE DA actually asks — and it is not what you expect

This is the important part. GATE DA does *not* ask the linear algebra of a first-year engineering course. Looking at every Linear Algebra question in the official 2024 and 2025 papers, four habits stand out:

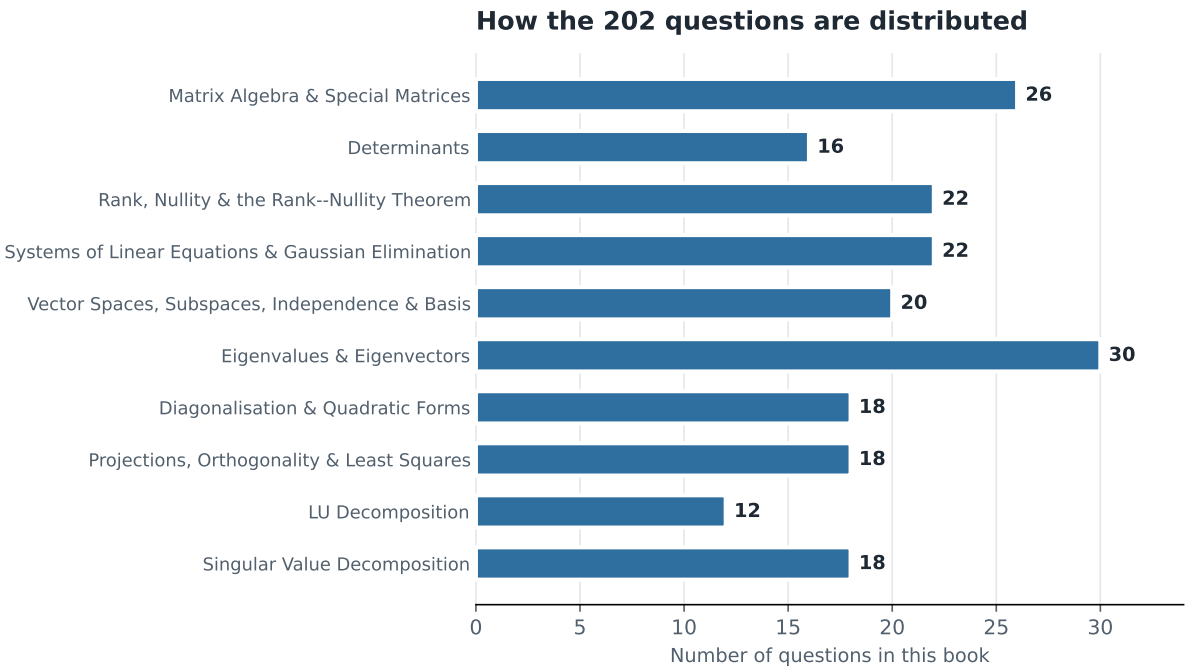
1. **Structure beats computation.** You are far more likely to be told "$A^3 = A$" or "$\|Ax\|_2 = \|x\|_2$" and asked what follows, than to be handed a 3×3 matrix and told to grind out its eigenvalues. Learn to read a *condition* and immediately write down what it forces.

2. **Rank-one matrices appear constantly.** xx^T, uu^T, $\sum_i x_i x_i^\mathsf{T}$ — these showed up in 2024 *and* twice in 2025. Know cold that xx^T has rank 1, is symmetric, and has a single non-zero eigenvalue equal to $x^\mathsf{T}x$.

3. **Matrix *types* are examined as properties.** Orthogonal, idempotent, projection, symmetric, Gram — usually as MSQs asking which statements hold. The entries never matter; the defining identity does.

4. **There is heavy overlap with Machine Learning.** Gram matrices, quadratic forms as decision boundaries, PCA and SVD. Time spent here is paid twice.

The number of questions given to each topic below follows that reality: eigenvalues and matrix properties get the most space, and pure hand-computation topics such as determinants get less than a conventional textbook would give them.



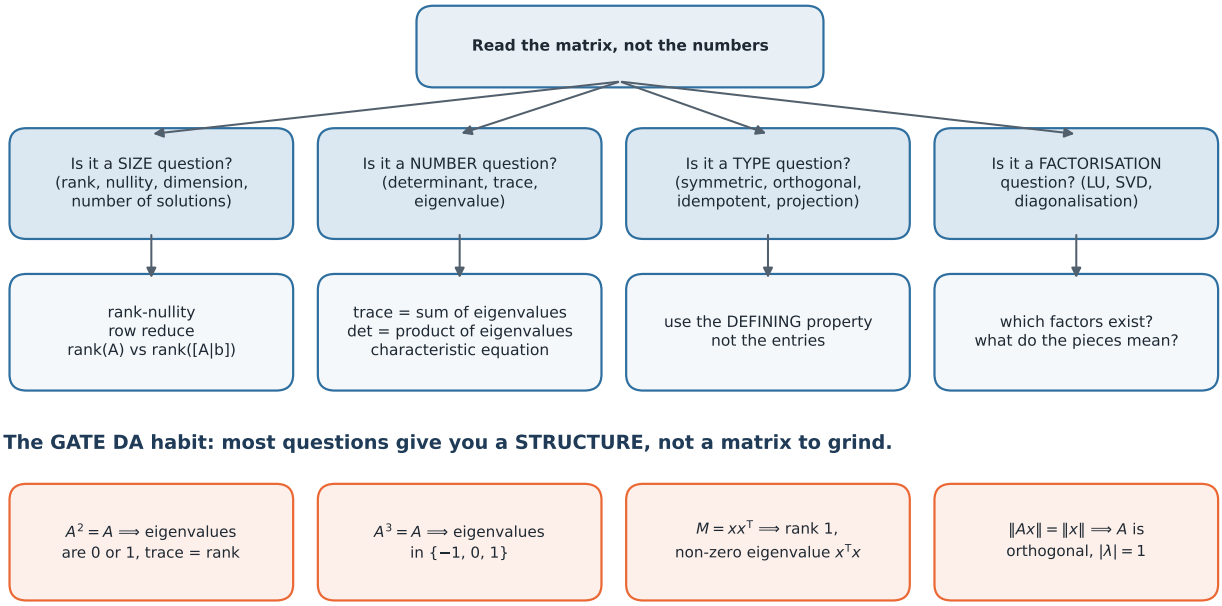
Question types and marking (same as the real paper)

Type	Description	Negative marking
MCQ	four options, exactly one correct	$-1/3$ (1-mark) or $-2/3$ (2-mark)
MSQ	four options, one *or more* correct; no partial credit	none
NAT	numerical answer, entered on a virtual keypad	none

This bank is 42% MCQ, 14% MSQ and 44% NAT. The real paper runs closer to 50/20/30; the extra NAT weighting here is deliberate, because NAT carries no negative marking and computing a rank or a determinant by hand is exactly the drill you want before the exam.

The habit that makes this subject fast

What is the question really asking about?



The GATE DA habit: most questions give you a STRUCTURE, not a matrix to grind.

Twelve facts that answer a disproportionate number of questions.

- trace = sum of eigenvalues • determinant = product of eigenvalues.
- $A^2 = A$ (idempotent) $\Rightarrow$ eigenvalues are 0 or 1, and $\operatorname{trace}(A) = \operatorname{rank}(A)$.
- $A^3 = A \Rightarrow$ every eigenvalue lies in $\{-1, 0, 1\}$.
- $M = xx^{\mathsf{T}} \Rightarrow$ rank $= 1$, symmetric, single non-zero eigenvalue $x^{\mathsf{T}}x$.
- $A = I + xx^{\mathsf{T}}$ with $\|x\| = 1 \Rightarrow$ eigenvalues are 2 (once) and 1 ($n - 1$ times).
- Q orthogonal $\Rightarrow Q^{-1} = Q^{\mathsf{T}}$, $\|Qx\| = \|x\|$, $|\lambda| = 1$, $\det Q = \pm 1$.
- Projection $P \Rightarrow P^2 = P$, $P^{\mathsf{T}} = P$, $\operatorname{trace}(P) = \operatorname{rank}(P) =$ dim of the subspace.
- rank + nullity = number of *columns*.
- Consistency: $\operatorname{rank}(A) = \operatorname{rank}([A \,|\, b])$; unique solution needs that common rank to equal n.
- Singular values = square roots of the eigenvalues of $A^{\mathsf{T}}A$; they are never negative.
- $\|A\|_F^2 = \sum \sigma_i^2$ and $\|A\|_2 = \sigma_{\max}$.
- X with independent columns $\Rightarrow X^{\mathsf{T}}X$ is invertible and positive definite.

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Answer Key **37**

1. Matrix Algebra & Special Matrices

26 questions • 13 MCQ, 5 MSQ, 8 NAT • Suggested time: 66 minutes

Q.1 [MCQ • 1 Mark • Easy]

If A is a 3×4 matrix and B is a 4×2 matrix, then the order of AB is

(A) 3×4 (B) 3×2 (C) 2×3 (D) 4×4

Q.2 [NAT • 1 Mark • Easy]

The trace of $\begin{bmatrix} 2 & 1 & 0 \\ 3 & 5 & 1 \\ 0 & 2 & 4 \end{bmatrix}$ is _____.

Q.3 [MCQ • 1 Mark • Easy]

For conformable matrices A and B, $(AB)^{\mathsf{T}}$ equals

(A) $B^{\mathsf{T}}A^{\mathsf{T}}$ (B) AB

(C) $A^{\mathsf{T}}B^{\mathsf{T}}$ (D) BA

Q.4 [MSQ • 1 Mark • Easy]

For an arbitrary square matrix A of order n, which of the following expression(s) is/are **always** defined?

(A) AA^{T} (B) A^{-1} (C) trace(A) (D) $A + A^{\mathsf{T}}$

Q.5 [MCQ • 1 Mark • Easy]

A square matrix A satisfying $A = A^{\mathsf{T}}$ is called

(A) symmetric (B) skew-symmetric (C) orthogonal (D) idempotent

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Q.6 [MCQ • 2 Marks • Moderate]

If A is a real skew-symmetric matrix of **odd** order, then $\det(A)$ equals

(A) -1 (B) it cannot be determined (C) 1 (D) 0

Q.7 [NAT • 2 Marks • Moderate]

For $A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$, the $(1,2)$ entry of A^5 is _____.

Q.8 [MSQ • 2 Marks • Moderate]

For square matrices A and B of the same order, which of the following statement(s) is/are **always** correct?

(A) $(A+B)^{\mathsf{T}} = A^{\mathsf{T}} + B^{\mathsf{T}}$ (B) $\text{trace}(AB) = \text{trace}(BA)$

(C) $\left(A^{\mathsf{T}}\right)^{\mathsf{T}} = A$ (D) $(AB)^{\mathsf{T}} = A^{\mathsf{T}}B^{\mathsf{T}}$

Q.9 [MCQ • 1 Mark • Moderate]

A matrix M is said to be **idempotent** if

(A) $M^2 = I$ (B) $M^{\mathsf{T}} = M^{-1}$ (C) $M^2 = M$ (D) $M^k = 0$ for some k

Q.10 [NAT • 2 Marks • Moderate]

A 5×5 idempotent matrix has rank 3. Its trace is _____.

Q.11 [MCQ • 1 Mark • Moderate]

If Q is a real orthogonal matrix, then Q^{-1} equals

(A) $\dfrac{Q}{\det Q}$ (B) Q (C) Q^{T} (D) $-Q$

Q.12 [NAT • 2 Marks • Moderate]

Let $x = (1,\ 2,\ 2)^{\mathsf{T}}$ and $M = xx^{\mathsf{T}}$. Then $\text{trace}(M) =$ _____.

Q.13 [MSQ • 2 Marks • Moderate]

Let $x \in \mathbb{R}^n$ be a non-zero vector and $M = xx^{\mathsf{T}}$ with $n \geq 2$. Which of the following statement(s) is/are correct?

(A) M is symmetric (B) $\text{rank}(M) = 1$

(C) M is invertible (D) The only non-zero eigenvalue of M is $x^{\mathsf{T}}x$

Q.14 [NAT • 2 Marks • Moderate]

A non-zero 3×3 matrix A satisfies $A^3 = 0$. Then $\text{trace}(A) =$ _____.

Q.15 [MCQ • 1 Mark • Moderate]

If A is an $n \times n$ real orthogonal matrix, then $|\det(A)|$ equals

(A) n (B) 1 (C) 0 (D) $\det(A^{\mathsf{T}})$ only

Q.16 [MCQ • 2 Marks • Hard]

For $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$, the value of $\text{trace}\big(A^2\big)$ is

(A) 29 (B) 25 (C) 14 (D) 22

Q.17 [MCQ • 2 Marks • Moderate]

Every real square matrix can be written **uniquely** as the sum of

(A) an orthogonal and a diagonal matrix (B) a symmetric and a skew-symmetric matrix

(C) two symmetric matrices (D) two idempotent matrices

Q.18 [NAT • 2 Marks • Moderate]

A 2×2 matrix has trace 5 and determinant 6. The value of $\text{trace}\left(A^2\right)$ is _____.

Q.19 [MCQ • 2 Marks • Moderate]

For square blocks A and B, the determinant of the block-diagonal matrix $\begin{bmatrix} A & 0 \\ 0 & B \end{bmatrix}$ equals

(A) $\det(A) - \det(B)$ (B) $\det(A)\det(B)$ (C) $\det(A) + \det(B)$ (D) $\det(AB) - 1$

Q.20 [MSQ • 2 Marks • Moderate]

Let M be an idempotent matrix. Which of the following statement(s) is/are correct?

(A) $\text{trace}(M) = \text{rank}(M)$

(B) Every eigenvalue of M is 0 or 1

(C) $I - M$ is idempotent

(D) M must be symmetric

Q.21 [NAT • 2 Marks • Hard]

Let $x \in \mathbb{R}^3$ with $x^{\mathsf{T}}x = 1$ and $A = I_3 + xx^{\mathsf{T}}$. Then $\det(A) =$ _____.

Q.22 [MCQ • 2 Marks • Hard]

Let $x \in \mathbb{R}^n$ with $\|x\|_2 = 1$ and $A = I_n + xx^{\mathsf{T}}$. The eigenvalues of A are

(A) 2 with multiplicity 1 and 1 with multiplicity $n - 1$

(B) 0 with multiplicity 1 and 1 with multiplicity $n - 1$

(C) 1 with multiplicity n

(D) 2 with multiplicity n

Q.23 [MCQ • 2 Marks • Moderate]

For $A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$, the value of $\text{trace}\left(A^{100}\right)$ is

(A) 1 (B) 0 (C) 100 (D) 2

Q.24 [MCQ • 2 Marks • Moderate]

If A and B are symmetric matrices of the same order, then AB is symmetric if and only if

(A) $A = B$

(B) $AB = I$

(C) A and B are both invertible

(D) $AB = BA$

Q.25 [MSQ • 2 Marks • Hard]

Let Q be a real orthogonal matrix. Which of the following statement(s) is/are correct?

(A) $\|Qx\|_2 = \|x\|_2$ for every x

(B) Every eigenvalue of Q has modulus 1

(C) $\det(Q) = 1$

(D) The columns of Q form an orthonormal set

Q.26 [NAT • 2 Marks • Moderate]

For $A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$, the value of $\text{trace}\left(A^{\mathsf{T}}A\right)$ is _____.

2. Determinants

16 questions • 9 MCQ, 2 MSQ, 5 NAT • Suggested time: 36 minutes

Q.1 [NAT • 1 Mark • Easy]

The determinant of $\begin{bmatrix} 3 & 1 \\ 2 & 4 \end{bmatrix}$ is _____.

Q.2 [MCQ • 1 Mark • Easy]

The determinant of $\begin{bmatrix} 2 & 5 & 7 \\ 0 & 3 & 1 \\ 0 & 0 & 4 \end{bmatrix}$ is

(A) 24 (B) 9 (C) 14 (D) 0

Q.3 [MCQ • 1 Mark • Easy]

If two rows of a square matrix are identical, its determinant is

(A) 1 (B) equal to the trace (C) 0 (D) -1

Q.4 [MSQ • 1 Mark • Easy]

Which of the following statement(s) about determinants is/are **always** correct?

(A) $\det(kA) = k\det(A)$ for a scalar k

(B) Interchanging two rows changes the sign of the determinant

(C) Adding a multiple of one row to another leaves the determinant unchanged

(D) $\det(A^{\mathsf{T}}) = \det(A)$

Q.5 [MCQ • 2 Marks • Moderate]

If A is a 3×3 matrix with $\det(A) = 5$, then $\det(2A)$ equals

(A) 40 (B) 5 (C) 20 (D) 10

Q.6 [MCQ • 1 Mark • Moderate]

If $\det(A) = 4$ for an invertible matrix A, then $\det(A^{-1})$ equals

(A) 0.25 (B) -4 (C) 1 (D) 4

Q.7 [MCQ • 1 Mark • Moderate]

If $\det(A) = 3$ and $\det(B) = -2$ for square matrices of the same order, then $\det(AB)$ equals

(A) 6 (B) 1 (C) -6 (D) -1

Q.8 [NAT • 2 Marks • Moderate]

The determinant of $\begin{bmatrix} 1 & 2 & 0 \\ 3 & -1 & 2 \\ 2 & 0 & 1 \end{bmatrix}$ is _____.

Q.9 [NAT • 2 Marks • Moderate]

A 3×3 matrix has eigenvalues 1, 2 and 5. Its determinant is _____.

Q.10 [NAT • 2 Marks • Hard]

For $M = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$, the value of $\det\left(M^2 + 12M\right)$ is _____.

Q.11 [MCQ • 1 Mark • Moderate]

If $\det(A) = 0$ for a square matrix A, then

(A) every eigenvalue of A is zero

(B) A is symmetric

(C) A is not invertible

(D) A is the zero matrix

Q.12 [MCQ • 2 Marks • Hard]

The determinant of the $n \times n$ matrix all of whose entries equal 1, for $n \geq 2$, is

(A) n^n (B) n (C) 1 (D) 0

Q.13 [MCQ • 2 Marks • Moderate]

The determinant of $\begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{bmatrix}$ is

(A) -2 (B) 2 (C) 0 (D) 1

Q.14 [MSQ • 2 Marks • Hard]

For an $n \times n$ matrix A, which of the following statement(s) is/are correct?

(A) $\det(A)$ equals the product of the eigenvalues of A

(B) $\det(A + B) = \det(A) + \det(B)$

(C) $\text{trace}(A)$ equals the sum of the eigenvalues of A

(D) $\det\left(A^k\right) = (\det A)^k$ for a positive integer k

Q.15 [MCQ • 2 Marks • Moderate]

If A is a 4×4 real orthogonal matrix, then $\det(A)$ is

(A) $+1$ or -1 (B) 4 (C) 0 (D) $+1$ only

Q.16 [NAT • 1 Mark • Easy]

The determinant of $\begin{bmatrix} 2 & 0 & 0 \\ 1 & 3 & 0 \\ 4 & 5 & 6 \end{bmatrix}$ is _____.

3. Rank, Nullity & the Rank–Nullity Theorem

22 questions • 8 MCQ, 3 MSQ, 11 NAT • Suggested time: 55 minutes

Q.1 [MCQ • 1 Mark • Easy]

The rank of $\begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$ is

(A) 4 (B) 1 (C) 2 (D) 0

Q.2 [NAT • 1 Mark • Easy]

The rank of the 3×3 matrix all of whose entries equal 1 is _____.

Q.3 [MCQ • 1 Mark • Easy]

For a matrix A of order $m \times n$, the rank–nullity theorem states that $\text{rank}(A) + \text{nullity}(A)$ equals

(A) n (B) $m + n$ (C) m (D) $\min(m, n)$

Q.4 [NAT • 1 Mark • Easy]

A 5×7 matrix has rank 4. Its nullity is _____.

Q.5 [NAT • 2 Marks • Moderate]

The rank of $\begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \\ 1 & 1 & 1 \end{bmatrix}$ is _____.

Q.6 [MCQ • 1 Mark • Moderate]

A 4×4 matrix has rank 2. The dimension of its null space is

(A) 3 (B) 2 (C) 1 (D) 0

Q.7 [MSQ • 2 Marks • Moderate]

For matrices A and B of compatible orders, which of the following statement(s) is/are **always** correct?

(A) $\text{rank}(AB) = \text{rank}(A) \cdot \text{rank}(B)$

(B) $\text{rank}(A) = \text{rank}\left(A^{\mathsf{T}}\right)$

(C) $\text{rank}(AB) \leq \min\{\text{rank}(A), \text{rank}(B)\}$

(D) $\text{rank}(A + B) \leq \text{rank}(A) + \text{rank}(B)$

Q.8 [NAT • 2 Marks • Moderate]

For non-zero vectors $x \in \mathbb{R}^m$ and $y \in \mathbb{R}^n$, the rank of xy^{T} is _____.

Q.9 [MCQ • 1 Mark • Moderate]

An $n \times n$ matrix A has rank n. It follows that

(A) A is orthogonal

(B) every eigenvalue of A equals 1

(C) A is symmetric

(D) A is invertible

Q.10 [NAT • 2 Marks • Moderate]

The rank of $\begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 2 \end{bmatrix}$ is _____.

Q.11 [NAT • 2 Marks • Hard]

Let A be a non-zero 4×4 real matrix in which the entries of every row sum to 0. The largest possible value of $\text{rank}(A)$ is _____.

Q.12 [MCQ • 1 Mark • Moderate]

The nullity of the 3×4 zero matrix is

(A) 4 (B) 0 (C) 1 (D) 3

Q.13 [NAT • 2 Marks • Moderate]

A 3×5 matrix has rank 3. The dimension of its column space is _____.

Q.14 [MSQ • 2 Marks • Hard]

For a real matrix A of order $m \times n$, which of the following statement(s) is/are correct?

(A) The null space of A is a subspace of $\mathbb{R}^m$

(B) The row space and the column space have the same dimension

(C) The column space of A is a subspace of $\mathbb{R}^m$

(D) The null space of A is a subspace of $\mathbb{R}^n$

Q.15 [NAT • 2 Marks • Moderate]

The nullity of $\begin{bmatrix} 2 & 1 \\ 4 & 2 \end{bmatrix}$ is _____.

Q.16 [MCQ • 2 Marks • Hard]

Let A be an $m \times n$ real matrix with $m < n$. Then the homogeneous system $Ax = 0$

(A) has exactly $n - m$ solutions

(B) always has infinitely many solutions

(C) has no solution

(D) has only the trivial solution

Q.17 [NAT • 2 Marks • Moderate]

The rank of $\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$ is _____.

Q.18 [NAT • 2 Marks • Hard]

A 6×4 real matrix has rank 4. Its nullity is _____.

Q.19 [MCQ • 2 Marks • Moderate]

For the system $Ax = b$ in n unknowns, if $\text{rank}(A) = \text{rank}([A \,|\, b]) < n$, then the system has

(A) exactly one solution (B) exactly n solutions (C) no solution (D) infinitely many solutions

Q.20 [MSQ • 2 Marks • Hard]

For an $n \times n$ real matrix A, which of the following statement(s) is/are correct?

(A) $\text{rank}(A^2) \leq \text{rank}(A)$

(B) $\text{rank}(A) = n$ if and only if $\det(A) \neq 0$

(C) $\text{rank}(A) = 0$ if and only if A is the zero matrix

(D) $\text{rank}(A) = \text{trace}(A)$

Q.21 [NAT • 2 Marks • Hard]

Let J_4 be the 4×4 matrix of all ones and $A = I_4 - \frac{1}{4}J_4$. Then $\text{rank}(A) =$ _____.

Q.22 [MCQ • 2 Marks • Hard]

Let X be an $n \times k$ real matrix ($n \geq k$) whose columns are linearly independent. The Gram matrix $X^{\mathsf{T}}X$ is

(A) invertible and positive definite (B) always the identity matrix

(C) skew-symmetric (D) singular

4. Systems of Linear Equations & Gaussian Elimination

22 questions • 10 MCQ, 3 MSQ, 9 NAT • Suggested time: 57 minutes

Q.1 [MCQ • 1 Mark • Easy]

A homogeneous system of linear equations $Ax = 0$

(A) always has at least the trivial solution

(B) never has a solution

(C) always has infinitely many solutions

(D) has a unique solution only if A is singular

Q.2 [NAT • 1 Mark • Easy]

For the system $x + y = 5$, $x - y = 1$, the value of x is _____.

Q.3 [MCQ • 1 Mark • Easy]

The system $Ax = b$ is consistent if and only if

(A) $b = 0$

(B) $\text{rank}(A) = \text{rank}([A \mid b])$

(C) A is square

(D) $\det(A) \neq 0$

Q.4 [NAT • 2 Marks • Moderate]

The system $kx + y = 1$, $2x + 2y = 3$ has **no** solution when $k =$ _____.

Q.5 [MCQ • 1 Mark • Moderate]

If $\text{rank}(A) = \text{rank}([A \mid b]) = r$ and the system has n unknowns with $r < n$, then the number of free variables is

(A) $n - r$ (B) n (C) 0 (D) r

Q.6 [NAT • 2 Marks • Moderate]

For the system $x + 2y + 3z = 6$, $2x + 4y + 6z = 12$, $x - y + z = 1$, the number of free variables is _____.

Q.7 [MSQ • 2 Marks • Moderate]

Which of the following statement(s) about Gaussian elimination is/are correct?

(A) Elementary row operations do not change the solution set

(B) Elementary row operations do not change the determinant

(C) Partial pivoting improves numerical stability

(D) Elementary row operations do not change the rank of the matrix

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Q.8 [NAT • 2 Marks • Hard]

The system $x + y + z = 6$, $x + 2y + 3z = 10$, $x + 2y + kz = 12$ fails to have a unique solution when $k =$ _____.

Q.9 [MCQ • 1 Mark • Moderate]

Cramer's rule can be applied to solve $Ax = b$ only when

(A) A has more rows than columns

(B) A is symmetric

(C) A is square and $\det(A) \neq 0$

(D) $b = 0$

Q.10 [NAT • 2 Marks • Moderate]

For $A = \begin{bmatrix} 2 & 1 \\ 1 & 3 \end{bmatrix}$ and $b = \begin{bmatrix} 5 \\ 10 \end{bmatrix}$, the value of y in the solution of $Ax = b$, where $x = (x, y)^{\mathsf{T}}$, is _____.

Q.11 [MCQ • 2 Marks • Hard]

Let A be a 3×5 real matrix. If the system $Ax = b$ is consistent, then it has

(A) at most two solutions

(B) infinitely many solutions

(C) exactly one solution

(D) no solution

Q.12 [MCQ • 2 Marks • Moderate]

Once a system has been reduced to an upper triangular form, solving it by back substitution takes

(A) $\Theta(n)$ operations

(B) $\Theta(n^3)$ operations

(C) $\Theta(n^2)$ operations

(D) $\Theta(n^2 \log n)$ operations

Q.13 [MSQ • 2 Marks • Moderate]

For a system $Ax = b$, which of the following statement(s) is/are correct?

(A) If $\operatorname{rank}(A) < \operatorname{rank}([A \,|\, b])$ the system is inconsistent

(B) If $b = 0$ and $\det(A) = 0$ the system has infinitely many solutions

(C) A linear system can have exactly two distinct solutions

(D) If A is invertible the system has exactly one solution

Q.14 [NAT • 2 Marks • Moderate]

The system $x + y = 2$, $2x + 2y = \lambda$ is consistent when $\lambda =$ _____.

Q.15 [MCQ • 2 Marks • Hard]

For an overdetermined system $Ax = b$ with more equations than unknowns and no exact solution, the standard remedy is to

(A) delete equations until the system becomes square

(B) solve the least-squares problem $\min \|Ax - b\|_2$

(C) conclude that A is singular

(D) set the extra variables to zero

Q.16 [NAT • 2 Marks • Moderate]

For the system $2x + 3y = 8$, $x - y = 1$, the value of x is _____ (rounded off to two decimal places).

Q.17 [MCQ • 2 Marks • Moderate]

Performing an elementary row operation on a matrix A is equivalent to

(A) post-multiplying A by an elementary matrix (B) transposing A

(C) multiplying A by its inverse (D) pre-multiplying A by an elementary matrix

Q.18 [NAT • 2 Marks • Hard]

A homogeneous system in 4 unknowns has coefficient matrix of rank 2. The dimension of its solution space is _____.

Q.19 [MCQ • 1 Mark • Moderate]

A homogeneous system of 3 linear equations in 4 unknowns

(A) has exactly one solution (B) always has a non-trivial solution

(C) is always inconsistent (D) has only the trivial solution

Q.20 [MSQ • 2 Marks • Hard]

Which of the following statement(s) about the effect of row operations on the determinant is/are correct?

(A) Interchanging two rows multiplies the determinant by -1

(B) Multiplying a single row by c multiplies the determinant by c^n

(C) Multiplying a single row by c multiplies the determinant by c

(D) Adding a multiple of one row to another leaves the determinant unchanged

Q.21 [NAT • 2 Marks • Moderate]

The rank of the augmented matrix $\left[\begin{array}{cc|c} 1 & 2 & 3 \\ 2 & 4 & 6 \end{array}\right]$ is _____.

Q.22 [MCQ • 2 Marks • Hard]

Solving a dense $n \times n$ linear system by Gaussian elimination requires

(A) $\Theta(n^3)$ arithmetic operations (B) $\Theta(n \log n)$ arithmetic operations

(C) $\Theta(2^n)$ arithmetic operations (D) $\Theta(n^2)$ arithmetic operations

5. Vector Spaces, Subspaces, Independence & Basis

20 questions • 8 MCQ, 3 MSQ, 9 NAT • Suggested time: 49 minutes

Q.1 [MCQ • 1 Mark • Easy]

Every subspace of a vector space must contain

(A) only unit vectors

(B) the zero vector

(C) a basis of the whole space

(D) at least two non-zero vectors

Q.2 [NAT • 1 Mark • Easy]

The dimension of $\mathbb{R}^5$ as a real vector space is _____.

Q.3 [MCQ • 1 Mark • Easy]

Which of the following is a subspace of $\mathbb{R}^3$?

(A) $\{(x, y, z) : x + y + z = 0\}$

(B) $\{(x, y, z) : x \geq 0\}$

(C) $\{(x, y, z) : x + y + z = 1\}$

(D) $\{(x, y, z) : x^2 + y^2 + z^2 = 1\}$

Q.4 [NAT • 1 Mark • Easy]

The dimension of the vector space of all 2×2 real matrices is _____.

Q.5 [MSQ • 2 Marks • Moderate]

Which of the following set(s) is/are subspaces of $\mathbb{R}^3$?

(A) $\{\alpha v : \alpha \in \mathbb{R}\}$ for a fixed non-zero $v \in \mathbb{R}^3$

(B) $\{x \in \mathbb{R}^3 : x_1 = x_2 = x_3\}$

(C) $\{x \in \mathbb{R}^3 : x_1 + x_2 + x_3 = 0\}$

(D) $\{x \in \mathbb{R}^3 : x_1 + x_2 + x_3 = 1\}$

Q.6 [MCQ • 1 Mark • Moderate]

The vectors $(1, 2)$ and $(2, 4)$ in $\mathbb{R}^2$ are

(A) linearly dependent

(B) orthogonal

(C) linearly independent

(D) a basis of $\mathbb{R}^2$

Q.7 [NAT • 2 Marks • Moderate]

The dimension of the space of all 3×3 real **symmetric** matrices is _____.

Q.8 [NAT • 2 Marks • Moderate]

The dimension of the space of all 3×3 real **skew-symmetric** matrices is _____.

Q.9 [MCQ • 1 Mark • Moderate]

Any set of four vectors in $\mathbb{R}^3$ is

(A) linearly independent (B) orthogonal

(C) a basis of $\mathbb{R}^3$ (D) linearly dependent

Q.10 [NAT • 2 Marks • Moderate]

The dimension of the span of $(1, 0, 1)$, $(0, 1, 1)$ and $(1, 1, 2)$ in $\mathbb{R}^3$ is _____.

Q.11 [MSQ • 2 Marks • Hard]

For vectors in $\mathbb{R}^n$, which of the following statement(s) is/are correct?

(A) Any n linearly independent vectors in $\mathbb{R}^n$ form a basis

(B) Any set containing the zero vector is linearly dependent

(C) Any $n + 1$ vectors in $\mathbb{R}^n$ are linearly dependent

(D) A linearly independent set in $\mathbb{R}^n$ may contain more than n vectors

Q.12 [MCQ • 1 Mark • Moderate]

The dimension of the vector space of all real polynomials of degree at most 3 is

(A) 4 (B) infinite (C) 3 (D) 5

Q.13 [NAT • 2 Marks • Moderate]

The dimension of the null space of the 1×3 matrix $\begin{bmatrix} 1 & 1 & 1 \end{bmatrix}$ is _____.

Q.14 [NAT • 2 Marks • Hard]

Let U and W be subspaces of $\mathbb{R}^7$ with $\dim U = 4$ and $\dim W = 5$. The smallest possible value of $\dim(U \cap W)$ is _____.

Q.15 [MCQ • 2 Marks • Moderate]

The set of all $n \times n$ invertible real matrices, under the usual matrix addition and scalar multiplication, is

(A) a vector space of dimension n

(B) a subspace of the space of symmetric matrices

(C) a vector space of dimension n^2

(D) not a vector space, because it does not contain the zero matrix

Q.16 [NAT • 2 Marks • Moderate]

Three non-zero, pairwise orthogonal vectors in $\mathbb{R}^4$ span a subspace of dimension _____.

Q.17 [MCQ • 2 Marks • Hard]

The dimension of the column space of a matrix is equal to

(A) its nullity (B) its trace (C) the number of its columns (D) its rank

---

Q.18 [NAT • 2 Marks • Moderate]

The dimension of the space of all 3×3 real **upper triangular** matrices is _____.

Q.19 [MSQ • 2 Marks • Hard]

Which of the following statement(s) about a basis is/are correct?

(A) $\mathbb{R}^n$ has infinitely many different bases for $n \geq 1$

(B) A basis is a linearly independent spanning set

(C) Every spanning set is a basis

(D) Every basis of a finite-dimensional space has the same number of vectors

---

Q.20 [MCQ • 2 Marks • Moderate]

If a set of three vectors spans $\mathbb{R}^3$, then the set

(A) must be orthogonal (B) is a basis of $\mathbb{R}^3$

(C) is linearly dependent (D) must contain the zero vector

---

6. Eigenvalues & Eigenvectors

30 questions • 12 MCQ, 3 MSQ, 15 NAT • Suggested time: 73 minutes

Q.1 [MCQ • 1 Mark • Easy]

The eigenvalues of $\begin{bmatrix} 2 & 0 \\ 0 & 5 \end{bmatrix}$ are

(A) 7 and 10 (B) 2 and 0 (C) 0 and 7 (D) 2 and 5

Q.2 [NAT • 1 Mark • Easy]

The sum of the eigenvalues of $\begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix}$ is _____.

Q.3 [MCQ • 1 Mark • Easy]

For a square matrix, the product of all its eigenvalues equals

(A) its determinant (B) its rank (C) its largest entry (D) its trace

Q.4 [NAT • 1 Mark • Easy]

The largest eigenvalue of $\begin{bmatrix} 5 & 4 \\ 1 & 2 \end{bmatrix}$ is _____.

Q.5 [MCQ • 1 Mark • Easy]

Every eigenvalue of a real symmetric matrix is

(A) real (B) positive (C) purely imaginary (D) an integer

Q.6 [NAT • 1 Mark • Moderate]

The smaller eigenvalue of $\begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$ is _____.

Q.7 [MCQ • 1 Mark • Moderate]

If λ is an eigenvalue of A, then the corresponding eigenvalue of A^2 is

(A) 2λ (B) λ^2 (C) λ (D) $\sqrt{\lambda}$

Q.8 [NAT • 1 Mark • Moderate]

If 2 is an eigenvalue of A, then the corresponding eigenvalue of $A + 3I$ is _____.

Q.9 [MSQ • 2 Marks • Moderate]

For square matrices A and B of the same order, which of the following statement(s) is/are correct?

(A) The eigenvalues of $A + B$ are the sums of the eigenvalues of A and B

(B) If A is invertible, the eigenvalues of A^{-1} are the reciprocals of those of A

(C) A and A^{T} have the same eigenvalues

(D) Similar matrices have the same eigenvalues

Q.10 [NAT • 1 Mark • Moderate]

A 3×3 matrix has eigenvalues 1, 2 and 3. Its determinant is _____.

Q.11 [MCQ • 2 Marks • Moderate]

The eigenvalues of an idempotent matrix are

(A) all zero (B) ± 1 only (C) all equal to the trace (D) 0 and 1 only

---

Q.12 [NAT • 2 Marks • Moderate]

For the rotation matrix $\begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$, the modulus of each eigenvalue is _____.

Q.13 [MCQ • 2 Marks • Hard]

If a square matrix A satisfies $A^3 = A$, then every eigenvalue of A lies in the set

(A) $\{-1, 0, 1\}$ (B) $\{0, 1\}$ (C) $\{0, 3\}$ (D) $\{-1, 1\}$

Q.14 [NAT • 2 Marks • Hard]

Let $x \in \mathbb{R}^3$ with $\|x\|_2 = 1$ and $A = I_3 + xx^{\mathsf{T}}$. The largest eigenvalue of A is _____.

Q.15 [MCQ • 2 Marks • Moderate]

Every eigenvalue of a nilpotent matrix is

(A) ± 1 (B) equal to the trace (C) 1 (D) 0

Q.16 [NAT • 2 Marks • Moderate]

A 2×2 invertible matrix has eigenvalues 2 and 4. Then $\text{trace}\left(A^{-1}\right)$ = _____ (rounded off to two decimal places).

Q.17 [MSQ • 2 Marks • Hard]

Let A be a real symmetric $n \times n$ matrix. Which of the following statement(s) is/are correct?

(A) A has n linearly independent eigenvectors

(B) All eigenvalues of A are positive

(C) A is orthogonally diagonalisable

(D) Eigenvectors corresponding to distinct eigenvalues are orthogonal

---

Q.18 [NAT • 2 Marks • Moderate]

For $A = \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$, the sum of the absolute values of the eigenvalues is _____.

Q.19 [MCQ • 2 Marks • Hard]

For $A = \begin{bmatrix} 3 & 1 \\ 0 & 3 \end{bmatrix}$, the geometric multiplicity of the eigenvalue 3 is

(A) 2 (B) 0 (C) 3 (D) 1

Q.20 [NAT • 2 Marks • Moderate]

The largest eigenvalue of $\begin{bmatrix} 6 & 2 \\ 2 & 3 \end{bmatrix}$ is _____.

Q.21 [MCQ • 2 Marks • Hard]

The Cayley–Hamilton theorem states that a square matrix

(A) is always diagonalisable

(B) commutes with its transpose

(C) has real eigenvalues

(D) satisfies its own characteristic equation

Q.22 [NAT • 2 Marks • Moderate]

A 2×2 matrix has trace 5 and determinant 6. Then $\text{trace}\left(A^{-1}\right) =$ _____ (rounded off to three decimal places).

Q.23 [NAT • 2 Marks • Hard]

The largest eigenvalue of the 4×4 matrix all of whose entries equal 1 is _____.

Q.24 [MCQ • 1 Mark • Moderate]

An eigenvector of a matrix

(A) may be the zero vector

(B) must have integer entries

(C) is by definition a non-zero vector

(D) must be a unit vector

Q.25 [MSQ • 2 Marks • Moderate]

For an $n \times n$ matrix A, which of the following statement(s) is/are correct?

(A) If A has n distinct eigenvalues then A is diagonalisable

(B) A diagonalisable matrix must have n distinct eigenvalues

(C) The sum of the eigenvalues of A equals $\text{trace}(A)$

(D) The eigenvalues of a triangular matrix are its diagonal entries

Q.26 [NAT • 2 Marks • Moderate]

The sum of the eigenvalues of $\begin{bmatrix} 2 & 0 & 0 \\ 1 & 3 & 0 \\ 4 & 5 & 4 \end{bmatrix}$ is _____.

Q.27 [MCQ • 2 Marks • Hard]

A real symmetric matrix is positive definite if and only if

(A) all its eigenvalues are strictly positive (B) its determinant is positive

(C) all its entries are positive (D) its trace is positive

Q.28 [NAT • 2 Marks • Hard]

For $A = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$, the value of $\text{trace}\left(A^{2024}\right)$ is _____.

Q.29 [MCQ • 1 Mark • Easy]

The characteristic polynomial of a 3×3 matrix is a polynomial of degree

(A) 9 (B) 3 (C) 6 (D) 2

Q.30 [NAT • 2 Marks • Hard]

A 3×3 matrix A satisfies $A^2 = A$ and $\text{rank}(A) = 2$. Then $\text{trace}(A) =$ _____.

7. Diagonalisation & Quadratic Forms

18 questions • 7 MCQ, 3 MSQ, 8 NAT • Suggested time: 45 minutes

Q.1 [MCQ • 1 Mark • Easy]

An $n \times n$ matrix is diagonalisable if and only if it has

(A) n linearly independent eigenvectors

(B) all eigenvalues non-zero

(C) orthonormal columns

(D) n distinct eigenvalues

Q.2 [NAT • 1 Mark • Easy]

The number of linearly independent eigenvectors of $\begin{bmatrix} 3 & 1 \\ 0 & 3 \end{bmatrix}$ is _____.

Q.3 [MCQ • 1 Mark • Moderate]

Every real symmetric matrix can be diagonalised by

(A) a nilpotent matrix

(B) a lower triangular matrix

(C) an orthogonal matrix

(D) a skew-symmetric matrix

Q.4 [NAT • 1 Mark • Moderate]

The determinant of the symmetric matrix of the quadratic form $Q(x) = 2x_1^2 + 3x_2^2$ is _____.

Q.5 [MCQ • 2 Marks • Moderate]

The quadratic form $Q(x) = x^{\mathsf{T}}Ax$ with $A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$ is

(A) indefinite

(B) positive definite

(C) negative definite

(D) positive semi-definite but not definite

Q.6 [NAT • 2 Marks • Moderate]

For the quadratic form with matrix $\begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix}$, the number of negative eigenvalues is _____.

Q.7 [MSQ • 2 Marks • Moderate]

A real symmetric matrix A is positive definite if and only if

(A) all its entries are strictly positive

(B) $x^{\mathsf{T}}Ax > 0$ for every non-zero x

(C) all its leading principal minors are strictly positive

(D) all its eigenvalues are strictly positive

Q.8 [NAT • 2 Marks • Hard]

For the quadratic form $Q(x_1, x_2) = x_1^2 + 4x_1x_2 + x_2^2$, the smaller eigenvalue of its symmetric matrix is _____.

Q.9 [MCQ • 2 Marks • Moderate]

If $A = PDP^{-1}$ with D diagonal, then A^k equals

(A) D^k (B) PD^kP^{-1} (C) P^kDP^{-k} (D) PDP^{-k}

Q.10 [NAT • 2 Marks • Hard]

For $A = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 2 \end{bmatrix}$, the geometric multiplicity of the eigenvalue 1 is _____.

Q.11 [MCQ • 1 Mark • Moderate]

In the standard treatment of a quadratic form $x^{\mathsf{T}}Ax$, the matrix A is always taken to be

(A) upper triangular (B) idempotent (C) orthogonal (D) symmetric

Q.12 [MSQ • 2 Marks • Hard]

Which of the following statement(s) about diagonalisation is/are correct?

(A) Every diagonalisable matrix is invertible

(B) Every real symmetric matrix is diagonalisable

(C) Similar matrices have the same characteristic polynomial

(D) A matrix with n distinct eigenvalues is diagonalisable

Q.13 [NAT • 2 Marks • Moderate]

For the quadratic form $Q(x_1, x_2) = 3x_1^2 - 2x_1x_2 + 3x_2^2$, the smallest eigenvalue of its symmetric matrix is _____.

Q.14 [MCQ • 2 Marks • Hard]

A real symmetric matrix is negative definite if and only if

(A) its determinant is negative (B) its trace is negative

(C) all its entries are negative (D) all its eigenvalues are strictly negative

Q.15 [NAT • 2 Marks • Moderate]

The number of independent entries in a 4×4 real symmetric matrix is _____.

Q.16 [MCQ • 2 Marks • Hard]

If A is real symmetric and positive definite, then A^{-1} is

(A) negative definite (B) also positive definite (C) indefinite (D) not necessarily symmetric

Q.17 [NAT • 1 Mark • Moderate]

For the quadratic form $Q(x) = x^{\mathsf{T}} I_2 x$ evaluated at $x = (3, 4)^{\mathsf{T}}$, the value of $Q(x)$ is _____.

Q.18 [MSQ • 2 Marks • Hard]

A real symmetric matrix has eigenvalues 5, 0 and -2. Which of the following statement(s) is/are correct?

(A) The matrix is singular

(B) The associated quadratic form is indefinite

(C) The rank of the matrix is 2

(D) The matrix is positive semi-definite

---

8. Projections, Orthogonality & Least Squares

18 questions • 6 MCQ, 3 MSQ, 9 NAT • Suggested time: 48 minutes

Q.1 [MCQ • 1 Mark • Easy]

Two non-zero vectors in $\mathbb{R}^n$ are orthogonal if and only if

(A) they have equal length

(B) their inner product is zero

(C) their sum is zero

(D) they are linearly dependent

Q.2 [NAT • 1 Mark • Easy]

The Euclidean norm of the vector $(3, 4)$ is _____.

Q.3 [MCQ • 1 Mark • Easy]

An orthogonal projection matrix P satisfies

(A) $P^{\mathsf{T}}P = I$

(B) $P^2 = P$ and $P^{\mathsf{T}} = P$

(C) $P^2 = 0$

(D) $P^2 = I$ and $P^{\mathsf{T}} = P$

Q.4 [NAT • 1 Mark • Moderate]

The length of the orthogonal projection of $(3, 4)$ onto the line spanned by $(1, 0)$ is _____.

Q.5 [MCQ • 2 Marks • Moderate]

For a non-zero vector $a \in \mathbb{R}^n$, the orthogonal projection matrix onto the line spanned by a is

(A) $a^{\mathsf{T}}a\, I$

(B) aa^{T}

(C) $\dfrac{aa^{\mathsf{T}}}{a^{\mathsf{T}}a}$

(D) $\dfrac{a^{\mathsf{T}}a}{aa^{\mathsf{T}}}$

Q.6 [NAT • 2 Marks • Moderate]

For $a = (1, 1)^{\mathsf{T}}$, the trace of the projection matrix $P = \dfrac{aa^{\mathsf{T}}}{a^{\mathsf{T}}a}$ is _____.

Q.7 [MSQ • 2 Marks • Moderate]

Let P be the orthogonal projection matrix onto a k-dimensional subspace of $\mathbb{R}^n$ with $k < n$. Which of the following statement(s) is/are correct?

(A) P is invertible (B) P is symmetric (C) $\operatorname{rank}(P) = k$ (D) $P^2 = P$

Q.8 [NAT • 2 Marks • Moderate]

The trace of the orthogonal projection matrix onto a 3-dimensional subspace of $\mathbb{R}^7$ is _____.

Q.9 [MCQ • 2 Marks • Hard]

The least-squares solution of $Ax = b$ satisfies

(A) $AA^{\mathsf{T}}x = b$ (B) $Ax = A^{\mathsf{T}}b$

(C) $x = A^{-1}b$ (D) $A^{\mathsf{T}}Ax = A^{\mathsf{T}}b$

Q.10 [NAT • 2 Marks • Moderate]

The angle between the vectors $(1, 0)$ and $(1, 1)$, in degrees, is _____.

Q.11 [MSQ • 2 Marks • Hard]

Which of the following statement(s) about the Gram–Schmidt process is/are correct?

(A) It produces an orthonormal set from a linearly independent set

(B) At every step the span of the vectors produced so far equals the span of the inputs used so far

(C) It requires the input vectors to be linearly independent

(D) Its output does not depend on the order of the input vectors

---

Q.12 [NAT • 2 Marks • Moderate]

Let Q be an $n \times 4$ real matrix with orthonormal columns. Then $\text{trace}\left(Q^{\mathsf{T}}Q\right) =$ _____.

Q.13 [MCQ • 2 Marks • Hard]

If the columns of a real matrix X are linearly independent, then $X^{\mathsf{T}}X$ is

(A) invertible and positive definite (B) singular

(C) always equal to the identity (D) skew-symmetric

Q.14 [NAT • 2 Marks • Moderate]

The perpendicular distance from the origin to the line $x + y = 2$ in $\mathbb{R}^2$ is _____ (rounded off to three decimal places).

Q.15 [NAT • 2 Marks • Hard]

For a non-zero $a \in \mathbb{R}^5$, the rank of $P = I_5 - \dfrac{aa^{\mathsf{T}}}{a^{\mathsf{T}}a}$ is _____.

Q.16 [MCQ • 2 Marks • Moderate]

If P is the orthogonal projection onto a subspace S, then $I - P$ is the orthogonal projection onto

(A) the whole space (B) the orthogonal complement of S

(C) the zero subspace (D) S itself

Q.17 [NAT • 2 Marks • Hard]

The least-squares fit of a constant c to the data values 2, 4 and 9 is $c =$ _____.

Q.18 [MSQ • 2 Marks • Moderate]

Which of the following statement(s) about an orthonormal set of vectors is/are correct?

(A) The set must span the whole space

(B) The set is linearly independent

(C) The inner product of any two distinct vectors in the set is zero

(D) Every vector in the set has unit length

---

9. LU Decomposition

12 questions • 4 MCQ, 2 MSQ, 6 NAT • Suggested time: 31 minutes

Q.1 [MCQ • 1 Mark • Easy]

In the Doolittle form of the decomposition $A = LU$,

(A) L and U are both symmetric

(B) both L and U are upper triangular

(C) L is lower triangular with unit diagonal and U is upper triangular

(D) L is orthogonal and U is diagonal

Q.2 [NAT • 1 Mark • Easy]

For $A = \begin{bmatrix} 2 & 1 \\ 4 & 3 \end{bmatrix}$ written as $A = LU$ with L unit lower triangular, the $(2,1)$ entry of L is _____.

Q.3 [NAT • 1 Mark • Moderate]

For the same $A = \begin{bmatrix} 2 & 1 \\ 4 & 3 \end{bmatrix}$, the $(2,2)$ entry of U is _____.

Q.4 [MCQ • 2 Marks • Moderate]

LU decomposition without row interchanges can fail when

(A) the matrix is symmetric

(B) the matrix has integer entries

(C) the matrix is positive definite

(D) a pivot encountered during elimination is zero

Q.5 [NAT • 2 Marks • Moderate]

For $A = \begin{bmatrix} 1 & 2 \\ 3 & 8 \end{bmatrix}$ written as $A = LU$ with L unit lower triangular, the $(2,2)$ entry of U is _____.

Q.6 [MSQ • 2 Marks • Moderate]

Which of the following statement(s) about the LU decomposition is/are correct?

(A) With L unit lower triangular, $\det(A)$ is the product of the diagonal entries of U

(B) $\det(A) = \det(L)\det(U)$

(C) Computing the decomposition costs $\Theta(n^3)$ operations

(D) Every square matrix admits an LU decomposition without row interchanges

Q.7 [NAT • 2 Marks • Hard]

Let $A = \begin{bmatrix} 2 & 4 & -2 \\ 4 & 9 & -3 \\ -2 & -3 & 7 \end{bmatrix} = LU$ with L unit lower triangular. Then $\det(A) =$ _____.

Q.8 [MCQ • 2 Marks • Moderate]

Once an LU decomposition of an $n \times n$ matrix is available, solving $Ax = b$ for a new right-hand side b costs

(A) $\Theta(n^3)$ operations

(B) $\Theta(n^3 \log n)$ operations

(C) $\Theta(n^2)$ operations

(D) $\Theta(n)$ operations

Q.9 [NAT • 2 Marks • Hard]

For the matrix $A = \begin{bmatrix} 2 & 4 & -2 \\ 4 & 9 & -3 \\ -2 & -3 & 7 \end{bmatrix}$ of the previous question, the $(3,3)$ entry of U is _____.

Q.10 [MCQ • 2 Marks • Moderate]

The factorisation written as $PA = LU$ is used when

(A) row interchanges (partial pivoting) are required

(B) the matrix is singular

(C) only the determinant is needed

(D) the matrix is orthogonal

Q.11 [NAT • 2 Marks • Hard]

For $A = \begin{bmatrix} 4 & 3 \\ 6 & 3 \end{bmatrix} = LU$ with L unit lower triangular, the product of the diagonal entries of U is _____.

Q.12 [MSQ • 2 Marks • Moderate]

Which of the following are reasons to prefer an LU decomposition over explicitly computing A^{-1} when solving linear systems?

(A) It requires fewer arithmetic operations

(B) It works even when A is singular

(C) It is numerically more stable

(D) The same factorisation can be reused for many right-hand sides

---

10. Singular Value Decomposition

18 questions • 7 MCQ, 2 MSQ, 9 NAT • Suggested time: 46 minutes

Q.1 [MCQ • 1 Mark • Easy]

Which of the following is true of the singular value decomposition $A = U\Sigma V^{\mathsf{T}}$?

(A) It exists only for invertible matrices

(B) It exists only for symmetric matrices

(C) It exists for every real $m \times n$ matrix

(D) It exists only for square matrices

Q.2 [NAT • 1 Mark • Easy]

The largest singular value of $\begin{bmatrix} 3 & 0 \\ 0 & 4 \end{bmatrix}$ is _____.

Q.3 [MCQ • 1 Mark • Easy]

Singular values of a real matrix are always

(A) complex numbers

(B) equal to the eigenvalues

(C) positive integers

(D) non-negative real numbers

Q.4 [NAT • 1 Mark • Moderate]

The sum of the singular values of $\begin{bmatrix} 3 & 0 \\ 0 & -4 \end{bmatrix}$ is _____.

Q.5 [MCQ • 2 Marks • Moderate]

The singular values of a real matrix A are

(A) the diagonal entries of A

(B) the eigenvalues of A

(C) the non-negative square roots of the eigenvalues of $A^{\mathsf{T}}A$

(D) the reciprocals of the eigenvalues of A

Q.6 [NAT • 2 Marks • Moderate]

Let $u = (1, 2, 2)^{\mathsf{T}}$ and $A = uu^{\mathsf{T}}$. The only non-zero singular value of A is _____.

Q.7 [MSQ • 2 Marks • Moderate]

For the decomposition $A = U\Sigma V^{\mathsf{T}}$, which of the following statement(s) is/are correct?

(A) $\|A\|_F^2$ equals the sum of the squares of the singular values

(B) $\operatorname{rank}(A)$ equals the number of non-zero singular values

(C) U and V are orthogonal matrices

(D) The singular values of A are the eigenvalues of A

Q.8 [NAT • 2 Marks • Moderate]

A 5×3 real matrix has rank 2. The number of its non-zero singular values is _____.

Q.9 [MCQ • 2 Marks • Hard]

For a real symmetric positive definite matrix, the singular values

(A) coincide with the eigenvalues

(B) are all equal to 1

(C) are the squares of the eigenvalues

(D) are the reciprocals of the eigenvalues

Q.10 [NAT • 2 Marks • Hard]

The largest singular value of $\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$ is _____ (rounded off to three decimal places).

Q.11 [MCQ • 2 Marks • Moderate]

The best rank-k approximation of a matrix in the Frobenius norm is obtained by

(A) keeping the k largest singular values and discarding the rest

(B) keeping the k smallest singular values

(C) rounding the entries of the matrix

(D) deleting all but k columns of the matrix

Q.12 [NAT • 2 Marks • Moderate]

Let $x_1, x_2, x_3 \in \mathbb{R}^5$ be orthonormal and $A = \sum_{i=1}^{3} x_i x_i^{\mathsf{T}}$. The sum of the singular values of A is _____.

Q.13 [MSQ • 2 Marks • Hard]

Comparing the SVD with the eigendecomposition, which statement(s) is/are correct?

(A) The eigendecomposition requires a square matrix

(B) The SVD exists for every real matrix, square or not

(C) For a real symmetric positive semi-definite matrix the two essentially coincide

(D) The SVD requires the matrix to be invertible

Q.14 [NAT • 1 Mark • Moderate]

The number of singular values of a 4×7 real matrix is _____.

Q.15 [MCQ • 2 Marks • Hard]

In principal component analysis of a mean-centred data matrix X, the principal directions are

(A) the rows of X with the largest norm

(B) the right singular vectors of X

(C) the columns of X with the largest variance

(D) the left singular vectors of X

Q.16 [NAT • 2 Marks • Hard]

A matrix has singular values 5, 3 and 1. Its Frobenius norm is _____ (rounded off to three decimal places).

Q.17 [MCQ • 2 Marks • Moderate]

The spectral norm $\|A\|_2$ of a real matrix equals

(A) its largest singular value

(B) its largest eigenvalue

(C) its Frobenius norm

(D) the sum of its singular values

Q.18 [NAT • 2 Marks • Hard]

A matrix has singular values 6, 2 and 0. The Frobenius-norm error of its best rank-one approximation is _____.

Answer Key

Answers only. A key with several letters (e.g. **ABC**) is an MSQ, where every listed option must be selected and no other. For NAT questions any value within the usual GATE tolerance of the figure shown is accepted.

1. Matrix Algebra & Special Matrices

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	B	**2**	11	**3**	A	**4**	ACD	**5**	A
6	D	**7**	5	**8**	ABC	**9**	C	**10**	3
11	C	**12**	9	**13**	ABD	**14**	0	**15**	B
16	A	**17**	B	**18**	13	**19**	B	**20**	ABC
21	2	**22**	A	**23**	D	**24**	D	**25**	ABD
26	285								

2. Determinants

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	10	**2**	A	**3**	C	**4**	BCD	**5**	A
6	A	**7**	C	**8**	1	**9**	10	**10**	-404
11	C	**12**	D	**13**	B	**14**	ACD	**15**	A
16	36								

3. Rank, Nullity & the Rank–Nullity Theorem

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	B	**2**	1	**3**	A	**4**	3	**5**	2
6	B	**7**	BCD	**8**	1	**9**	D	**10**	2
11	3	**12**	A	**13**	3	**14**	BCD	**15**	1
16	B	**17**	2	**18**	0	**19**	D	**20**	ABC
21	3	**22**	A						

4. Systems of Linear Equations & Gaussian Elimination

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	A	**2**	3	**3**	B	**4**	1	**5**	A
6	1	**7**	ACD	**8**	3	**9**	C	**10**	3
11	B	**12**	C	**13**	ABD	**14**	4	**15**	B
16	2.20	**17**	D	**18**	2	**19**	B	**20**	ACD
21	1	**22**	A						

5. Vector Spaces, Subspaces, Independence & Basis

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	B	**2**	5	**3**	A	**4**	4	**5**	ABC
6	A	**7**	6	**8**	3	**9**	D	**10**	2
11	ABC	**12**	A	**13**	2	**14**	2	**15**	D
16	3	**17**	D	**18**	6	**19**	ABD	**20**	B

6. Eigenvalues & Eigenvectors

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	D	**2**	7	**3**	A	**4**	6	**5**	A
6	1	**7**	B	**8**	5	**9**	BCD	**10**	6
11	D	**12**	1	**13**	A	**14**	2	**15**	D
16	0.75	**17**	ACD	**18**	4	**19**	D	**20**	7
21	D	**22**	0.833	**23**	4	**24**	C	**25**	ACD
26	9	**27**	A	**28**	2	**29**	B	**30**	2

7. Diagonalisation & Quadratic Forms

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	A	**2**	1	**3**	C	**4**	6	**5**	B
6	1	**7**	BCD	**8**	-1	**9**	B	**10**	1
11	D	**12**	BCD	**13**	2	**14**	D	**15**	10
16	B	**17**	25	**18**	ABC				

8. Projections, Orthogonality & Least Squares

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	B	**2**	5	**3**	B	**4**	3	**5**	C
6	1	**7**	BCD	**8**	3	**9**	D	**10**	45
11	ABC	**12**	4	**13**	A	**14**	1.414	**15**	4
16	B	**17**	5	**18**	BCD				

9. LU Decomposition

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	C	**2**	2	**3**	1	**4**	D	**5**	2
6	ABC	**7**	8	**8**	C	**9**	4	**10**	A
11	-6	**12**	ACD						

10. Singular Value Decomposition

Q	Ans	Q	Ans	Q	Ans	Q	Ans	Q	Ans
1	C	**2**	4	**3**	D	**4**	7	**5**	C
6	9	**7**	ABC	**8**	2	**9**	A	**10**	1.618
11	A	**12**	3	**13**	ABC	**14**	4	**15**	B
16	5.916	**17**	A	**18**	2				

Score sheet

Anything below 70% on a topic means the concept, not the practice, is the problem — go back and revise before re-attempting.

Topic	Questions	Correct	Time taken
Matrix Algebra & Special Matrices	26		
Determinants	16		
Rank, Nullity & the Rank–Nullity Theorem	22		
Systems of Linear Equations & Gaussian Elimination	22		
Vector Spaces, Subspaces, Independence & Basis	20		
Eigenvalues & Eigenvectors	30		
Diagonalisation & Quadratic Forms	18		
Projections, Orthogonality & Least Squares	18		
LU Decomposition	12		
Singular Value Decomposition	18		
Total	**202**		